NLP Part 6: Named Entity Recognition (NER)

What is Named Entity Recognition?

Named Entity Recognition (NER) is an NLP technique used to identify and classify important entities in text.

Common Entity Types

Entity	Meaning	Example
PERSON	Person's name	Steve Jobs
ORGANIZATION	Company or organization	Apple
LOCATION	Place or location	Finland
DATE	Date or year	2026
TIME	Time	10:30 AM
MONEY	Monetary value	$500

Example

"Apple was founded by Steve Jobs in California."

NER can identify:

- **Apple** → ORGANIZATION
- **Steve Jobs** → PERSON
- **California** → LOCATION

In [6]:
```
import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('maxent_ne_chunker_tab')
nltk.download('words')

print("NLTK is ready!")
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\himats\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\himats\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     C:\Users\himats\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package maxent_ne_chunker_tab to
[nltk_data]     C:\Users\himats\AppData\Roaming\nltk_data...
NLTK is ready!
[nltk_data]   Package maxent_ne_chunker_tab is already up-to-date!
[nltk_data] Downloading package words to
[nltk_data]     C:\Users\himats\AppData\Roaming\nltk_data...
[nltk_data]   Package words is already up-to-date!
```

1. Basic NER Example

NER normally works after:

Text → Tokenization → POS Tagging → Named Entity Recognition

First, we divide the sentence into tokens.

Then we assign POS tags.

Finally, NER identifies named entities.

In [7]:
```python
from nltk import word_tokenize, pos_tag, ne_chunk

text = "Barack Obama visited Finland."

# Tokenization
tokens = word_tokenize(text)

# POS Tagging
pos_tags = pos_tag(tokens)

# Named Entity Recognition
entities = ne_chunk(pos_tags)

print(entities)
```

```
(S
  (PERSON Barack/NNP)
  (PERSON Obama/NNP)
  visited/VBD
  (GPE Finland/NNP)
  ./.)
```

2. Extract Named Entities

Instead of displaying the complete NER tree, we can extract only the named entities.

For each entity, we want to display:

Entity Name → Entity Type

In [8]:
```
from nltk import word_tokenize, pos_tag, ne_chunk

text = """
Barack Obama visited Finland and met representatives
from Microsoft in Helsinki.
"""

tokens = word_tokenize(text)
pos_tags = pos_tag(tokens)
tree = ne_chunk(pos_tags)

for chunk in tree:

    if hasattr(chunk, 'label'):

        entity = " ".join(
            word for word, tag in chunk.leaves()
        )

        print(entity, "->", chunk.label())
```

```
Barack -> PERSON
Obama -> PERSON
Finland -> GPE
Microsoft -> ORGANIZATION
Helsinki -> GPE
```

What is GPE?

NLTK often uses the label **GPE**.

GPE = Geopolitical Entity

It represents geographical areas that also have political or administrative meaning.

Examples:

- Finland → GPE
- Pakistan → GPE
- Helsinki → GPE
- London → GPE

3. NER with More Text

Let's use a larger example containing:

- People
- Organizations

- Countries
- Cities

In [13]:
```
text = """
Elon Musk founded SpaceX in the United States.
microsoft was founded by Bill Gates and Paul Allen.
nokia is a Finnish company headquartered in Espoo, Finland.
"""

tokens = word_tokenize(text)
pos_tags = pos_tag(tokens)
tree = ne_chunk(pos_tags)

for chunk in tree:

    if hasattr(chunk, 'label'):

        entity = " ".join(
            word for word, tag in chunk.leaves()
        )

        print(entity, "->", chunk.label())
```

```
Elon -> PERSON
Musk -> PERSON
SpaceX -> ORGANIZATION
United States -> GPE
Bill Gates -> PERSON
Paul Allen -> PERSON
Finnish -> GPE
Espoo -> GPE
Finland -> GPE
```

NER Pipeline

The basic NLTK NER pipeline is:

Raw Text ↓ Tokenization ↓ POS Tagging ↓ NER ↓ Named Entities

Example

"Steve Jobs founded Apple."

↓

Tokens

["Steve", "Jobs", "founded", "Apple", "."]

↓

POS Tags

Steve → NNP
Jobs → NNP

founded → VBD
Apple → NNP

↓

NER

Steve Jobs → PERSON
Apple → ORGANIZATION

4. Store Different Entity Types

We can also store detected entities in separate lists.

For example:

- People
- Organizations
- Locations

In [14]:
```
text = """
Barack Obama visited Microsoft in the United States.
Bill Gates later travelled to Finland.
"""

tokens = word_tokenize(text)
pos_tags = pos_tag(tokens)
tree = ne_chunk(pos_tags)

people = []
organizations = []
locations = []

for chunk in tree:

    if hasattr(chunk, 'label'):

        entity = " ".join(
            word for word, tag in chunk.leaves()
        )

        label = chunk.label()

        if label == "PERSON":
            people.append(entity)

        elif label == "ORGANIZATION":
            organizations.append(entity)

        elif label in ["GPE", "LOCATION"]:
            locations.append(entity)

print("People:")
print(people)
```

```
print("\nOrganizations:")
print(organizations)

print("\nLocations:")
print(locations)
```

```
People:
['Barack', 'Obama', 'Microsoft', 'Bill Gates']

Organizations:
[]

Locations:
['United States', 'Finland']
```

Exercise: Named Entity Recognition

Use the following text:

"Tim Cook works at Apple in California. Satya Nadella is the CEO of Microsoft. Microsoft has offices in Finland."

Task

Use NER to find:

1. PERSON
2. ORGANIZATION
3. LOCATION / GPE

Then store them in three separate lists:

- People
- Organizations
- Locations

text = """

Tim Cook works at Apple in California. Satya Nadella is the CEO of Microsoft. Microsoft has offices in Finland. """

Step 1: Tokenize

tokens = _____

Step 2: POS Tagging

pos_tags = _____

Step 3: NER

entities = _____

print(entities)

Why is NER Useful?

NER helps computers automatically extract important information from large amounts of text.

Applications

- Extracting people's names from news articles
- Finding company names
- Finding countries and cities
- Extracting information from documents
- Search engines
- Question answering
- Information extraction
- Document analysis

Simple Example

"John works for Google in London."

NER can transform this unstructured sentence into structured information:

Entity	Type
John	PERSON
Google	ORGANIZATION
London	LOCATION

NER Summary

Named Entity Recognition (NER) identifies important entities in text.

Common Entities

- PERSON → Person
- ORGANIZATION → Company or organization

- GPE / LOCATION → Country, city, or place
- DATE → Date
- TIME → Time
- MONEY → Monetary value

Basic NLP Pipeline

Text → Tokenization → POS Tagging → NER

Easy Way to Remember

> **POS tagging tells us what a word does grammatically.**
>
> **NER tells us what real-world entity a word or phrase represents.**

Next Topics in Natural Language Processing

In this course, we have covered the basic NLP preprocessing concepts.

There are many more advanced NLP techniques that we will **not cover in detail here**.

Further NLP Topics

1. **Semantic Similarity**
   - Measuring how similar words, sentences, or documents are in meaning.
2. **Word Co-occurrence**
   - Studying which words frequently appear together.
3. **Bag of Words (BoW)**
   - Representing text using word occurrence or word frequency.
4. **TF-IDF**
   - Measuring how important a word is in a document compared with other documents.
5. **N-grams**
   - Analyzing sequences of words, such as bigrams and trigrams.
6. **Sentiment Analysis**
   - Identifying positive, negative, or neutral opinions in text.
7. **Text Classification**
   - Assigning text to predefined categories.
8. **Topic Modeling**

- Discovering important topics from a collection of documents.

9. **Word Embeddings**
    - Representing words as numerical vectors.
    - Examples: Word2Vec, GloVe, FastText
10. **Transformers**
    - Modern deep-learning models for understanding language.
    - Examples: BERT, RoBERTa
11. **Large Language Models (LLMs)**
    - Advanced models capable of understanding and generating natural language.

NLP Learning Path

Text Extraction

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Cleaning

↓

Tokenization

↓

Stopword Removal

↓

POS Tagging

↓

Normalization

↓

Stemming & Lemmatization

↓

WordNet

↓

Named Entity Recognition (NER)

↓

Advanced NLP

Semantic Similarity → TF-IDF → Sentiment Analysis →
Word Embeddings → Transformers → Large Language Models

Key Message

> We have learned the fundamental steps needed to prepare and understand text data.
> These concepts provide the foundation for more advanced NLP techniques.